$$S = 1 + (1+d)r + (1+2d)r^2 + \cdots + (1+nd)r^n + \cdots$$

$$S = (1 + r + r^2 + \cdots) + d(r + 2r^2 + 3r^3 + \cdots)$$

$$rS = (r + r^2 + r^3 + \cdots) + d(r^2 + 2r^3 + 3r^4 + \cdots)$$

$$\because |r| < 1, \qquad \therefore 1 + r + r^2 + \cdots = \frac{1}{1-r}$$

$$\begin{aligned} S - rS &= 1 + d(r + r^2 + r^3 + \cdots) \\ &= 1 + d \cdot \frac{r}{1-r} \end{aligned}$$

$$S(1-r) = 1 + \cdot \frac{rd}{1-r}$$

$$\implies \boxed{S = \frac{1}{1-r} + \frac{rd}{(1-r)^2}}$$

$\mathbb{E}(x)$ is the expected number of shots Arthur takes to hit the target

$$\mathbb{E}(x) = a \cdot 1 + (1-a)(1 + \mathbb{E}(x)) = \frac{1}{a}$$

Arthur wins either immediately, with probability a, or after both players miss once, with probability $a'b'\alpha$

$$\alpha = a + a'b'\alpha \qquad \implies \boxed{\alpha = \frac{a}{1-a'b'}}$$

$$\beta = 1 - \alpha = \frac{a'b}{1-a'b'}$$

If Arthur wins on his nth shot, then total shots $= n + (n-1) = 2n - 1$

If Boadicea wins on her nth shot, then total shots $= n + n = 2n$

$$\mathbb{E}(x) = \sum_{n=1}^{\infty} (2n-1)(a'b')^{n-1}a + \sum_{n=1}^{\infty} 2n(a'b')^{n-1}a'b$$

$$\begin{aligned} \mathbb{E}(x) &= a\sum_{n=1}^{\infty} (2n-1)(a'b')^{n-1} + a'b\sum_{n=1}^{\infty} 2n(a'b')^{n-1} \\ &= a\left(2\sum_{n=1}^{\infty} n(a'b')^{n-1} - \sum_{n=1}^{\infty} (a'b')^{n-1}\right) + 2a'b\sum_{n=1}^{\infty} n(a'b')^{n-1} \end{aligned}$$

$$\sum_{n=1}^{\infty}(a'b')^{n-1} = \frac{1}{1-a'b'}, \qquad \sum_{n=1}^{\infty} n(a'b')^{n-1} = \frac{1}{(1-a'b')^2}$$

$$\mathbb{E}(x) = a\left(\frac{2}{(1-a'b')^2} - \frac{1}{1-a'b'}\right) + 2a'b \cdot \frac{1}{(1-a'b')^2} = \frac{a+a'b}{(1-a'b')^2}$$

$$\because a + a'b = a(1+b') = a(2-b) \qquad \therefore \mathbb{E}(x) = \frac{a(2-b)}{(1-a'b')^2}$$

$$\alpha = \frac{a}{1-a'b'}, \qquad \beta = \frac{a'b}{1-a'b'}$$

$$\frac{\alpha}{a} + \frac{\beta}{b} = \frac{1}{1-a'b'} + \frac{a'}{1-a'b'} = \frac{2-a}{1-a'b'}$$

$$\implies \boxed{E = \frac{\alpha}{a} + \frac{\beta}{b}}$$

This solution was submitted by **Jon**.

Published and refined by **Jon**.

Last updated: April 15, 2026